Elijah E. Mullens

Department of Astronomy and Carl Sagan Institute, Cornell University,
122 Sciences Drive, Ithaca, NY 14853, USA
eem85@cornell.edu, elijah-mullens.github.io

Education

Ph.D. Astronomy, *Cornell University* — August 2027 (estimated)
Advisor: Prof. Nikole lewis (nikole.lewis@cornell.edu)
Dissertation Title: "Cloudy With a Chance of Silicates:
An Exploration of Exoplanet Aerosols and Weather"

M.S. Astronomy, *Cornell University* — November 2024

B.S. Mathematics, Physics, and Astrophysics, Cum Laude, *University of Florida* — May 2021
Minor in East Asian Languages and Literatures - Japanese

A.S. *Tallahassee Community College / Florida State University Associate of Arts* — May 2017

Honors and Awards

Cranson W. and Edna B. Shelly Outstanding Teaching Award, *Cornell Department of Astronomy*	2025
NASA Summer Internship Program	2025
NSF Graduate Research Fellowship	2023-2027
Graduate School Dean's Excellence Fellowship, *Cornell University*	2022-2023
JASSO Scholarship, *Nagoya University*	2019-2020
Florida Bright Futures Scholarship - Academic Scholars Award	2017-2021
Presidential Scholarship, *University of Florida*	2017-2021
Dean's List, President's Honor Roll, *University of Florida*	2017-2021

Relevant Experience

NASA Goddard Space Flight Center — Summer 2025
Intern (PI: Dr. Sarah Moran)
Project: "Experimental and Modeling Investigation of Exoplanet Cloud Properties"

Cornell University, Department of Astronomy — 2024-2025
Teaching Assistant/Head Teaching Assistant
ASTRO 1195 (Observational Astronomy)
Smaller class (30 students). Held night labs, graded homework, and gave guest lectures.
ASTRO 1102 (Our Solar System)
Large class (130 students). Head-TA with multiple duties such as: attending lecture, holding hours, developing material for discussion section, organizing TA duties

Space Telescope Science Institute (STScI) — Summer 2022
Space Astronomy Summer Program Intern (PI: Dr. Catherine Zucker)
Project: "Unveiling the Nature of Diffuse Interstellar Envelopes Around Dense Star-forming Clouds"
Developed a pipeline to compare interstellar cloud simulations to observations, gave a presentation at

the 2022 SASP Symposium, will give a poster presentation at AAS 241 and Protostars and Planets VII. Resulting paper accepted for publication in ApJ.

Challenger Learning Center of Tallahassee 2021-2022

Planetarium Instructor

Created planetarium shows using Digistar3, ran planetarium shows, prepared and presented live the monthly 'Monthly Skies Over Tallahassee' and other educational pre-shows ('Monthly Astronomy Learning Topic', 'Space News', 'Monthly Solar System Object').

Meitetsu Work Travels Inc. - Nagoya University 2019-2020

English Camp Tutor

Led and organized group activities for over 1,000 children. Gave presentation on home country and taught basic mathematics.

Nagoya University, Department of Physics 2019-2020

Visiting Undergraduate Researcher

Further topics in Statistical Physics with Professor John Wojdylo, covering microcanonical and canonical formalism, quantum fluids, and mean field theory.

Nagoya University, Department of Earth and Environmental Sciences 2019-2020

Visiting Undergraduate Researcher

Further topics in Earth and Planetary Sciences with Professor Marc Humblet, research paper written on TESS and planetary detection methods.

University of Florida, Department of Astronomy Fall 2020

Undergraduate Researcher

Contributed to Professor Charles Telesco's IMPS (Integrated Miniature Polarimeter and Spectrograph) project. Achieved an understanding of the instrument while learning further topics on polarimetry and astrobiology.

University of Florida, Department of Mathematics Fall 2019

Undergraduate Researcher

Differential Geometry project with Professor Luca Di Cerbo covering Do Carmo's Differential Geometry of Curves and Surfaces, with a presentation at the end.

Certifications

Florida State University

Global Partners Certificate (2021)

Equity, Diversity, and Inclusion Certificate (2021)

Coursera

The University of Sydney: Data-driven Astronomy (2021)

NUPACE Program

Nagoya University: Certificate of Completion (2020)

Approved Grants & Observing Proposals

JWST, Cycle 5, 6.8 hours (Co-I, PI Boehm, JWST-GO-12237) 2026-2029

"A Definitive Transmission Survey of the Atmosphere of 55 Cnc e"

JWST, Cycle 5, 9.8 hours (Co-I, PI Lothringer, JWST-GO-10290) 2026-2029

"Mistaken Identity? Resolving Ultra-hot Jupiter NUV Absorption to Measure Refractory-to-Volatile Ratio"

JWST, Cycle 4, 13.7 hours (Co-I, PI MacDonald, JWST-GO-09033) 2025-2028
"By the Ashes of Stars: A Chemical Census of a White Dwarf Planet"

JWST, Cycle 4, 176.4 hours (Co-I, PI Faherty, JWST-GO-08441) 2025-2028
"The Cold Worlds Spectral Library"

JWST, Cycle 4, 37.7 hours (**PI**, JWST-GO-08309) 2025-2028
"It's (poly)Morphin' Time! Solving the quartz quandry of WASP-17b"

JWST, Cycle 4, 46.1 hours (Co-I, PI Moran, JWST-GO-07849) 2025-2028
"Meeting their Match: Measuring Cloudy Hot Jupiter Counterparts to Understand the Silicate Cloud Regime with JWST/MIRI"

JWST, Cycle 4, 31.7 hours (**PI**, JWST-GO-07686) 2025-2028
"Brown Dwarf Broiler: Probing Chemical Quenching and Heat Redistribution in a Highly-Eccentric Brown Dwarf"

NSF Graduate Research Fellowship ($159,000, Advisor: Prof. Nikole Lewis) 2023-2027
"Chemical Timescales in the Atmospheres of Highly Eccentric Sub-Neptunes"

Other Worlds Laboratory (OWL) ($2000) Summer 2025
In support of participation in OWL summer school

Cornell Graduate School Research Travel Grant ($2000) Summer 2025
In support of internship at NASA Goddard Space Flight Center

Cornell Graduate School Conference Grant ($700) Winter 2024
In support of presentation at 243rd American Astronomy Society Meeting

Cornell Graduate School Conference Grant ($700) Spring 2023
In support of presentation at Protostars & Planets VII

Presentations (Talks & Posters)

BOWIE+ Collaboration, Invited Speaker February 2026
"Head in the (Silicate) Clouds: Silicate Sundogs and Grain Alignment"

Modern Language Association (MLA), Accepted Talk January 2026
"An idea you'll have to give birth to': An exploration of astronomy and alternative relationality in Planet-Size X-Men"

American Geophysical Union (AGU), Accepted Talk December 2025
"HEL-Fire and Brimstone: High Temperature Optical Properties of Quartz and Tridymite for Substellar Cloud Studies"

American Astronomical Society (AAS246), Accepted iPoster June 2025
"Silicate Sundogs and Sandy Thunderstorms: Probing the Effects of Grain Directionality and Chemistry in Exoplanet Observations"

American Astronomical Society (AAS246), LabAstro Workshop Lightning Talk June 2025
"The Quartz Quandary of WASP-17b"

Emerging Researchers in Exoplanet Science (ERESIX), Accepted Talk July 2024
"A New Atmospheric Retrieval Paradigm for Exoplanet Aerosols"

Outer Planets Assessment Group (OPAG), Lightning Talk+Poster June 2024

"Life After Death: Europa in the Habitable Zone of a Red Sun and White Dwarf"

Astrobiology Graduate Conference (AbGradCon), Accepted Poster — June 2024
"Life After Death: Europa in the Habitable Zone of a Red Sun and White Dwarf"

American Astronomical Society (AAS243), Accepted iPoster — January 2024
"A New Atmospheric Retrieval Paradigm for Exoplanet Aerosols"

American Astronomical Society (AAS241), Accepted iPoster — January 2023
"Unveiling the Nature of Diffuse Interstellar Envelopes Around Dense Star-forming Clouds"

Protostars and Planets 7, Accepted Poster — April 2023
"Unveiling the Nature of Diffuse Interstellar Envelopes Around Dense Star-forming Clouds"

Professional Service & Associations

Graduate Mentor (Isabela Huckabee)	2024-present
Undergraduate Mentor (Ian Branigan)	2024-present
AbGradCon 2024, *SOC Member*	2023-2024
Florida State University (FSU) Astronomy Club, *Founder*	2022
Astronomers for Planet Earth, *Member*	2021-2022
oSTEM Global Chapter, *Member*	2021-2022
Circle at Nagoya University "Ametama", *Member*	2019-2020
Nagoya University Language Exchange Club, *Member*	2019-2020
Nagoya University Foreign Students Club, *Member*	2019-2020
oSTEM University of Florida, *Member*	2018-2019
University of Florida Japanese Language and Culture Club, *Member*	2017-2018
University of Florida Model UN (GatorMUN), *Member*	2017-2018

Public Outreach

Cornell University Department of Astronomy Outreach, *Volunteer*	2023-present
Graduate Student School Outreach Program (GRASHOPR), *Volunteer*	2023-2024
Sea-to-See Marine Biology Outreach Program, *Volunteer*	Summer 2022
Challenger Learning Center of Tallahassee, *Volunteer*	2021-2022
Tallahassee Astronomical Society, *Volunteer*	2021-2022
Tallahassee Scientific Society, *Event Organizer*	2021-2022

Selected Media

"Moment of Science: "Sun dogs" are possible on other planets, new study suggests", *ABC News*, 8/28/2025
"Sun dogs, other celestial effects could appear in alien skies", *Cornell Chronicle*, 7/30/2025
"Cornell astronomers win time on James Webb Space Telescope", *Cornell Chronicle*, 6/17/2025
"When the sun dies, could life survive on the Jupiter ocean moon Europa?", *Space.com*, 6/03/2025
"Will Europa Become a New Habitable World When the Sun Becomes a Red Giant?", *Universe Today*, 5/27/2025

Publications

–6 first author manuscripts

Submitted/In preparation

E. Mullens et al. (2026) "Rocky Planet Picture Show: The Rocky Planet Picture Show: Implementation of Surface Reflection and Emission in POSEIDON with Application to and Interpretation of JWST Data". *Submitted to PSJ*

E. Mullens et al. (2026) "HELFire and Brimstone: Lab-Measured, Temperature-Dependent Optical Properties of Particulate Quartz, Tridymite, and Amorphous Silica in Transmission with an Application to Substellar Cloud Studies", *In Prep*

Refereed Journal Articles

[14] **Mullens, Elijah** and N. K. Lewis. Silicate Sundogs: Probing the Effects of Grain Directionality in Exoplanet Observations. ApJ, 988(2):L43, Aug. 2025, 2508.01839.

[13] **Mullens, Elijah**, B. Schmidt, L. Kaltenegger, and N. K. Lewis. Life after death: Europa in the evolving habitable zone of a Red Sun. MNRAS, 540(1):1329–1344, June 2025, 2505.15495.

[12] K. S. Sotzen, J. Lustig-Yaeger, K. B. Stevenson, S.-M. Tsai, R. C. Challener, J. Goyal, N. K. Lewis, D. R. Louie, L. C. Mayorga, D. Valentine, H. R. Wakeford, L. Alderson, N. H. Allen, T. J. Fauchez, A. Glidden, A. Gressier, S. M. Hörst, J. Huang, Z. Lin, A. M. Mandell, **Mullens, Elijah**, S. Peacock, E. W. Schwieterman, J. A. Valenti, C. M. Mountain, M. Perrin, R. P. van der Marel, Consortium On Habitability, and Atmospheres Of M-Dwarf Planets (Champs). Non-detection of Nightside Emission from HAT-P-26b due to Stellar Contamination. *Research Notes of the American Astronomical Society*, 9(12):352, Dec. 2025.

[11] W. W. Meynardie, M. R. Meyer, R. J. MacDonald, P. Calissendorff, **Mullens, Elijah**, G. M. Zarazua, A. Roy, H. Ganta, E. C. Gonzales, A. Adams, N. Lewis, Y. Hong, and J. Lunine. Ross 458 C: Gas Giant or Brown Dwarf? ApJ, 994(2):237, Dec. 2025, 2509.22803.

[10] J. Lustig-Yaeger, K. S. Sotzen, K. B. Stevenson, S.-M. Tsai, R. C. Challener, J. Goyal, N. K. Lewis, D. R. Louie, L. C. Mayorga, D. Valentine, H. R. Wakeford, L. Alderson, N. H. Allen, T. J. Fauchez, A. Glidden, A. Gressier, S. M. Hörst, J. Huang, Z. Lin, A. M. Mandell, **Mullens, Elijah**, S. Peacock, E. W. Schwieterman, J. A. Valenti, C. M. Mountain, M. Perrin, and R. P. van der Marel. JWST-TST DREAMS: The Nightside Emission and Chemistry of WASP-17b. ApJ, 994(1):L4, Nov. 2025, 2510.06169.

[9] D. R. Louie, **Mullens, Elijah**, L. Alderson, A. Glidden, N. K. Lewis, H. R. Wakeford, N. E. Batalha, K. D. Colón, A. Gressier, D. Long, M. Radica, N. Espinoza, J. Goyal, R. J. MacDonald, E. M. May, S. Seager, K. B. Stevenson, J. A. Valenti, N. H. Allen, C. I. Cañas, R. C. Challener, D. Grant, J. Huang, Z. Lin, D. Valentine, M. Clampin, M. Perrin, L. Pueyo, R. P. van der Marel, and C. M. Mountain. JWST-TST DREAMS: A Precise Water Abundance for Hot Jupiter WASP-17b from the NIRISS SOSS Transmission Spectrum. AJ, 169(2):86, Feb. 2025, 2412.03675.

[8] A. Gressier, R. J. MacDonald, N. Espinoza, H. R. Wakeford, N. K. Lewis, J. Goyal, D. R. Louie, M. Radica, N. E. Batalha, D. Long, E. M. May, **Mullens, Elijah**, S. Seager, K. B. Stevenson, J. A.

Valenti, L. Alderson, N. H. Allen, C. I. Cañas, R. C. Challener, K. Colón, A. Glidden, D. Grant, J. Huang, Z. Lin, D. Valentine, C. M. Mountain, L. Pueyo, M. D. Perrin, and R. P. van der Marel. JWST-TST DREAMS: A Supersolar Metallicity in WASP-17 b's Dayside Atmosphere from NIRISS SOSS Eclipse Spectroscopy. AJ, 169(2):57, Feb. 2025, 2410.08149.

[7] A. Gressier, N. E. Batalha, N. Wogan, L. Alderson, D. Doud, N. Espinoza, R. J. MacDonald, H. R. Wakeford, J. A. Valenti, N. K. Lewis, S. Seager, K. B. Stevenson, N. H. Allen, C. I. Cañas, R. C. Challener, A. Glidden, J. Huang, Z. Lin, D. R. Louie, C. Maguire, **Mullens, Elijah**, K. Sotzen, D. Valentine, M. Clampin, L. Pueyo, R. P. van der Marel, and C. M. Mountain. JWST-TST DREAMS: Sulfur Dioxide in the Atmosphere of the Neptune-mass Planet HAT-P-26 b from NIRSpec G395H Transmission Spectroscopy. AJ, 170(5):292, Nov. 2025, 2509.16082.

[6] A. Glidden, S. Ranjan, S. Seager, N. Espinoza, R. J. MacDonald, N. H. Allen, C. I. Cañas, D. Grant, A. Gressier, K. B. Stevenson, N. E. Batalha, N. K. Lewis, D. Long, H. R. Wakeford, L. Alderson, R. C. Challener, K. Colón, J. Huang, Z. Lin, D. R. Louie, **Mullens, Elijah**, K. S. Sotzen, J. A. Valenti, D. Valentine, M. Clampin, C. M. Mountain, M. Perrin, and R. P. van der Marel. JWST-TST DREAMS: Secondary Atmosphere Constraints for the Habitable Zone Planet TRAPPIST-1 e. ApJ, 990(2):L53, Sept. 2025, 2509.05407.

[5] N. Espinoza, N. H. Allen, A. Glidden, N. K. Lewis, S. Seager, C. I. Cañas, D. Grant, A. Gressier, S. Courreges, K. B. Stevenson, S. Ranjan, K. Colón, B. M. Morris, R. J. MacDonald, D. Long, H. R. Wakeford, J. A. Valenti, L. Alderson, N. E. Batalha, R. C. Challener, J. Huang, Z. Lin, D. R. Louie, **Mullens, Elijah**, D. Valentine, C. M. Mountain, L. Pueyo, M. D. Perrin, A. Bellini, J. Kammerer, M. Libralato, I. Rebollido, E. Rickman, S. T. Sohn, and R. P. van der Marel. JWST-TST DREAMS: NIRSpec/PRISM Transmission Spectroscopy of the Habitable Zone Planet TRAPPIST-1 e. ApJ, 990(2):L52, Sept. 2025, 2509.05414.

[4] D. Valentine, H. R. Wakeford, R. C. Challener, N. E. Batalha, N. K. Lewis, D. Grant, **Mullens, Elijah**, L. Alderson, J. Goyal, R. J. MacDonald, E. M. May, S. Seager, K. B. Stevenson, J. A. Valenti, N. H. Allen, N. Espinoza, A. Glidden, A. Gressier, J. Huang, Z. Lin, D. Long, D. R. Louie, M. Clampin, M. Perrin, R. P. van der Marel, and C. M. Mountain. JWST-TST DREAMS: Nonuniform Dayside Emission for WASP-17b from MIRI/LRS. AJ, 168(3):123, Sept. 2024, 2410.08148.

[3] **Mullens, Elijah**, C. Zucker, C. E. Murray, and R. Smith. Characterizing the 3D Structure of Molecular Cloud Envelopes in the Cloud Factory Simulations. ApJ, 966(1):127, May 2024, 2403.03112.

[2] **Mullens, Elijah**, N. K. Lewis, and R. J. MacDonald. Implementation of Aerosol Mie Scattering in POSEIDON with Application to the Hot Jupiter HD 189733 b's Transmission, Emission, and Reflected Light Spectrum. ApJ, 977(1):105, Dec. 2024, 2410.19253.

[1] D. Grant, N. K. Lewis, H. R. Wakeford, N. E. Batalha, A. Glidden, J. Goyal, **Mullens, Elijah**, R. J. MacDonald, E. M. May, S. Seager, K. B. Stevenson, J. A. Valenti, C. Visscher, L. Alderson, N. H. Allen, C. I. Cañas, K. Colón, M. Clampin, N. Espinoza, A. Gressier, J. Huang, Z. Lin, D. Long, D. R. Louie, M. Peña-Guerrero, S. Ranjan, K. S. Sotzen, D. Valentine, J. Anderson, W. O. Balmer, A. Bellini, K. K. W. Hoch, J. Kammerer, M. Libralato, C. M. Mountain, M. D. Perrin, L. Pueyo, E. Rickman, I. Rebollido, S. T. Sohn, R. P. van der Marel, and L. L. Watkins. JWST-TST DREAMS: Quartz Clouds in the Atmosphere of WASP-17b. ApJ, 956(2):L32, Oct. 2023, 2310.08637.